Pandemic Mortality Analysis Report

📁 Dataset: `all_weekly_excess_deaths.csv`

Entries: 5,770 | **Regions:** Multiple countries
Time Span: Weekly data from late 2019 through the COVID-19 pandemic
Main Focus: Mortality trends (COVID, non-COVID, excess), normalized per 100k population.

📌 Dataset Overview

Column	Description
`country`, `region`, `region_code`	Geographical identifiers
`start_date`, `end_date`, `year`, `week`	Weekly time periods
`days`	Number of days in the week (usually 7)
`population`	Regional population
`total_deaths`	Total deaths recorded in that week
`expected_deaths`	Baseline/expected deaths without a pandemic
`covid_deaths`	Officially reported COVID-19 deaths
`non_covid_deaths`	Deaths not directly attributed to COVID-19
`excess_deaths`	Difference between actual and expected deaths
`covid_deaths_per_100k`, `excess_deaths_per_100k`	Normalized values per 100,000 people
`excess_deaths_pct_change`	Percent change from expected deaths

📊 Key Metrics Analyzed

1. 🦠 COVID-19 Deaths Trends

Tracks weekly changes in reported COVID-19 deaths. Useful to:

- Identify peak pandemic waves
- Compare severity across countries/regions

2. 📈 Excess Deaths

Computed as:

excess_deaths = total_deaths - expected_deaths

Helps measure:

- Unreported COVID deaths
- Indirect deaths (e.g., from overwhelmed healthcare systems)

3. 📉 Expected vs Actual Deaths

A direct comparison showing if regions are performing better or worse than historical norms.

4. 📦 Normalized Rates per 100k

- `covid_deaths_per_100k`
- `excess_deaths_per_100k`
 Standardizes comparisons across regions with different population sizes.

5. ⚖️ Percent Change in Excess Deaths

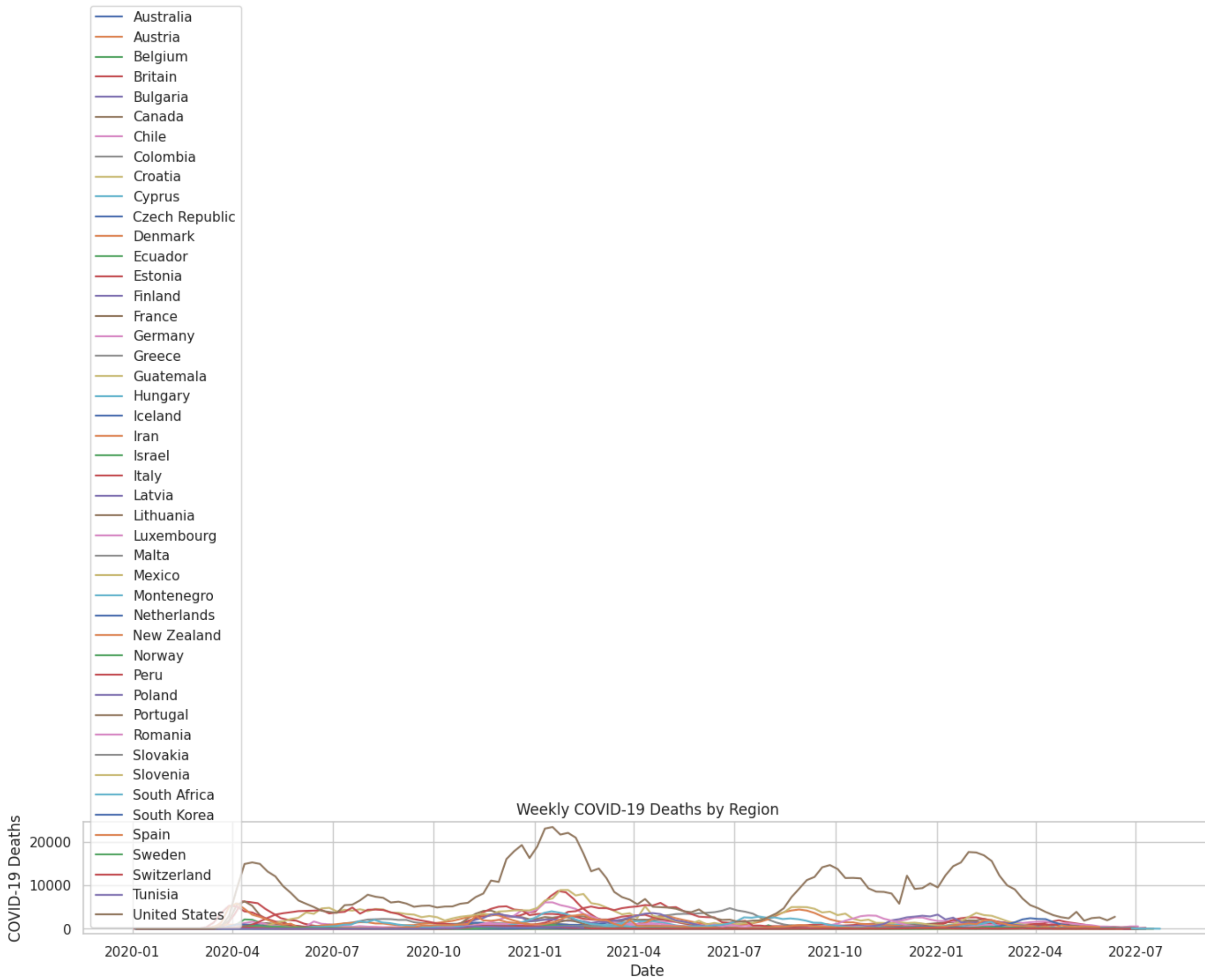
Shows volatility in death trends:

excess_deaths_pct_change = (excess_deaths / expected_deaths) * 100

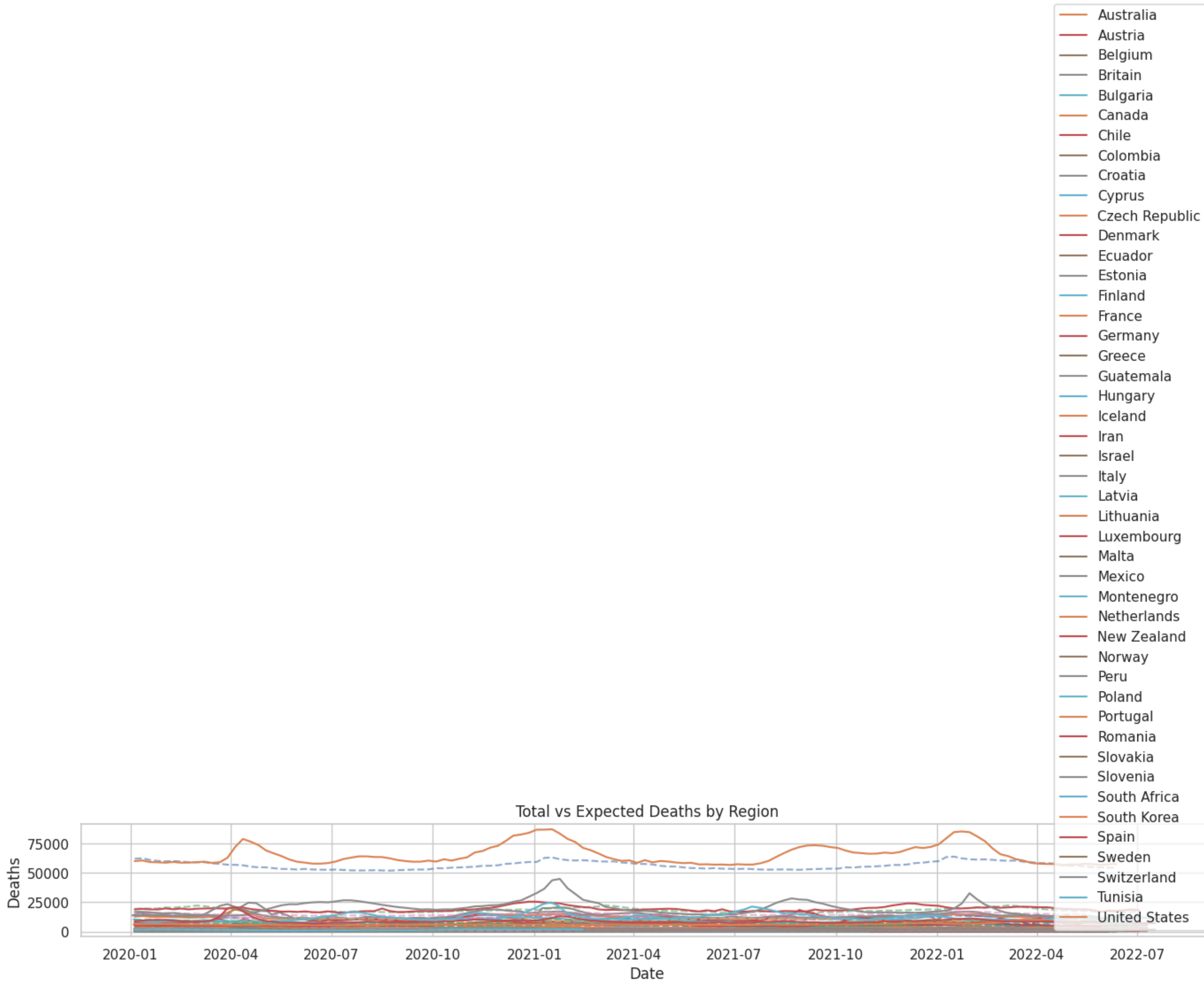
📈 Visual Analysis

Each of these insights will be accompanied by visualizations:

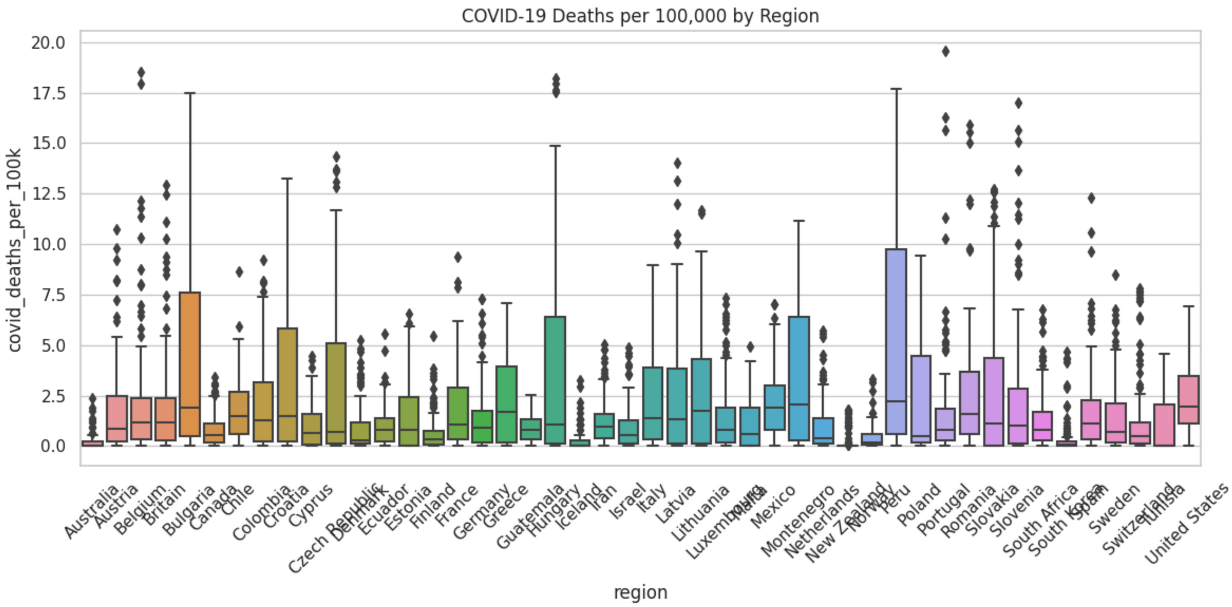
1. **Weekly COVID-19 Deaths by Region** – line chart of pandemic waves


Weekly COVID-19 Deaths by Region
Australia
Austria
Belgium
Britain
Bulgaria
Canada
Chile
Colombia
Croatia
Cyprus
Czech Republic
Denmark
Ecuador
Estonia
Finland
France
Germany
Greece
Guatemala
Hungary
Iceland
Iran
Israel
Italy
Latvia
Lithuania
Luxembourg
Malta
Mexico
Montenegro
Netherlands
New Zealand
Norway
Peru
Poland
Portugal
Romania
Slovakia
Slovenia
South Africa
South Korea
Spain
Sweden
Switzerland
Tunisia
United States
COVID-19 Deaths
20000
10000
0
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01
2022-04
2022-07
Date

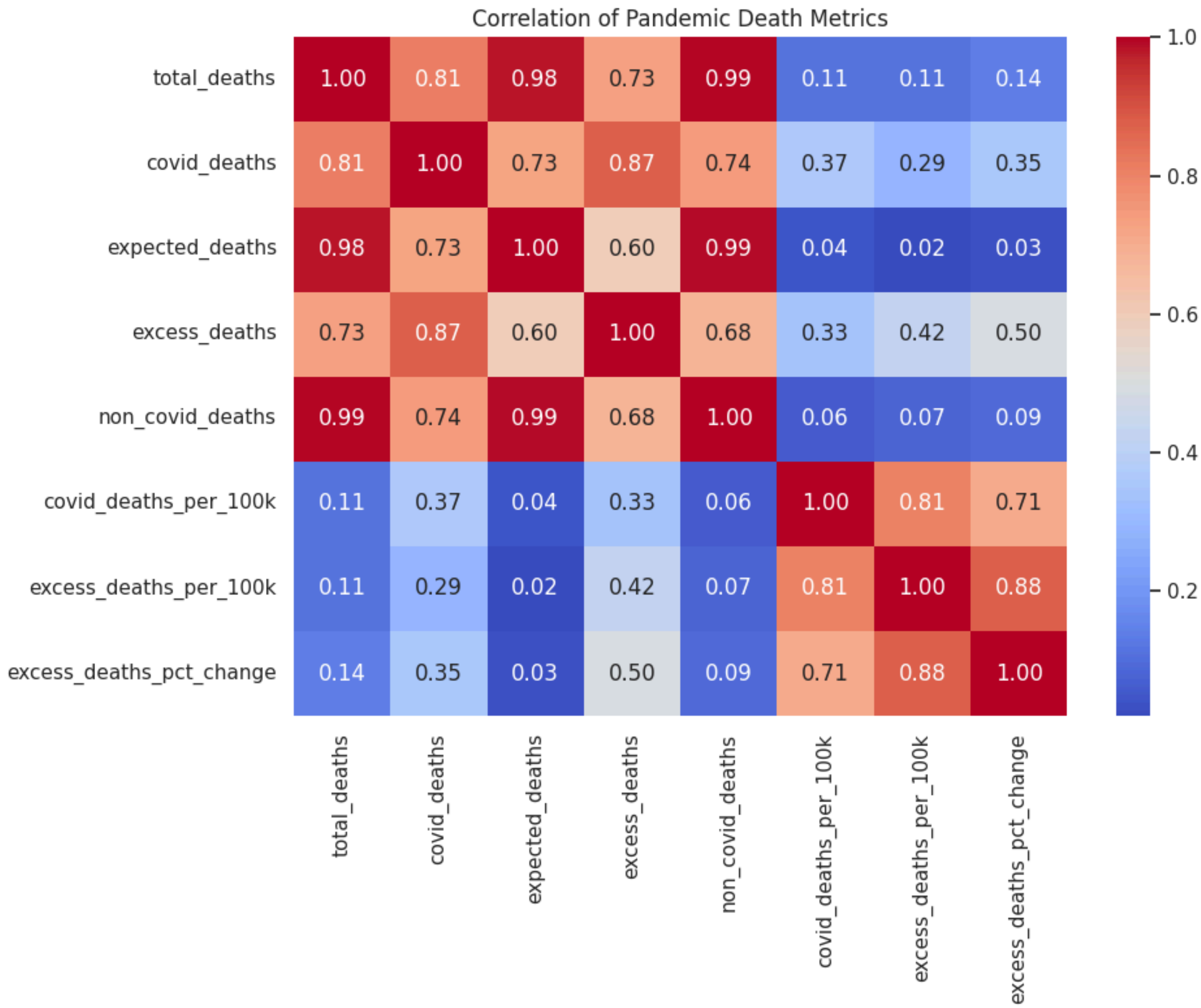
2. **Total vs Expected Deaths** – comparing weekly baseline with real



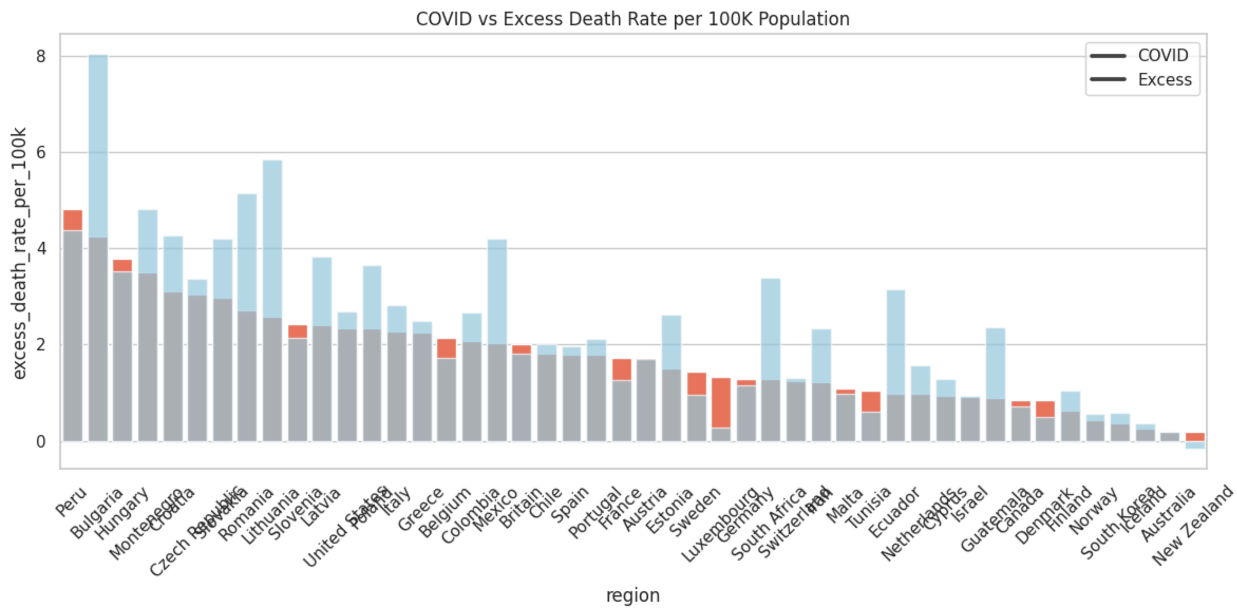
3. **COVID-19 Deaths per 100k** – regional disparities in mortality rates



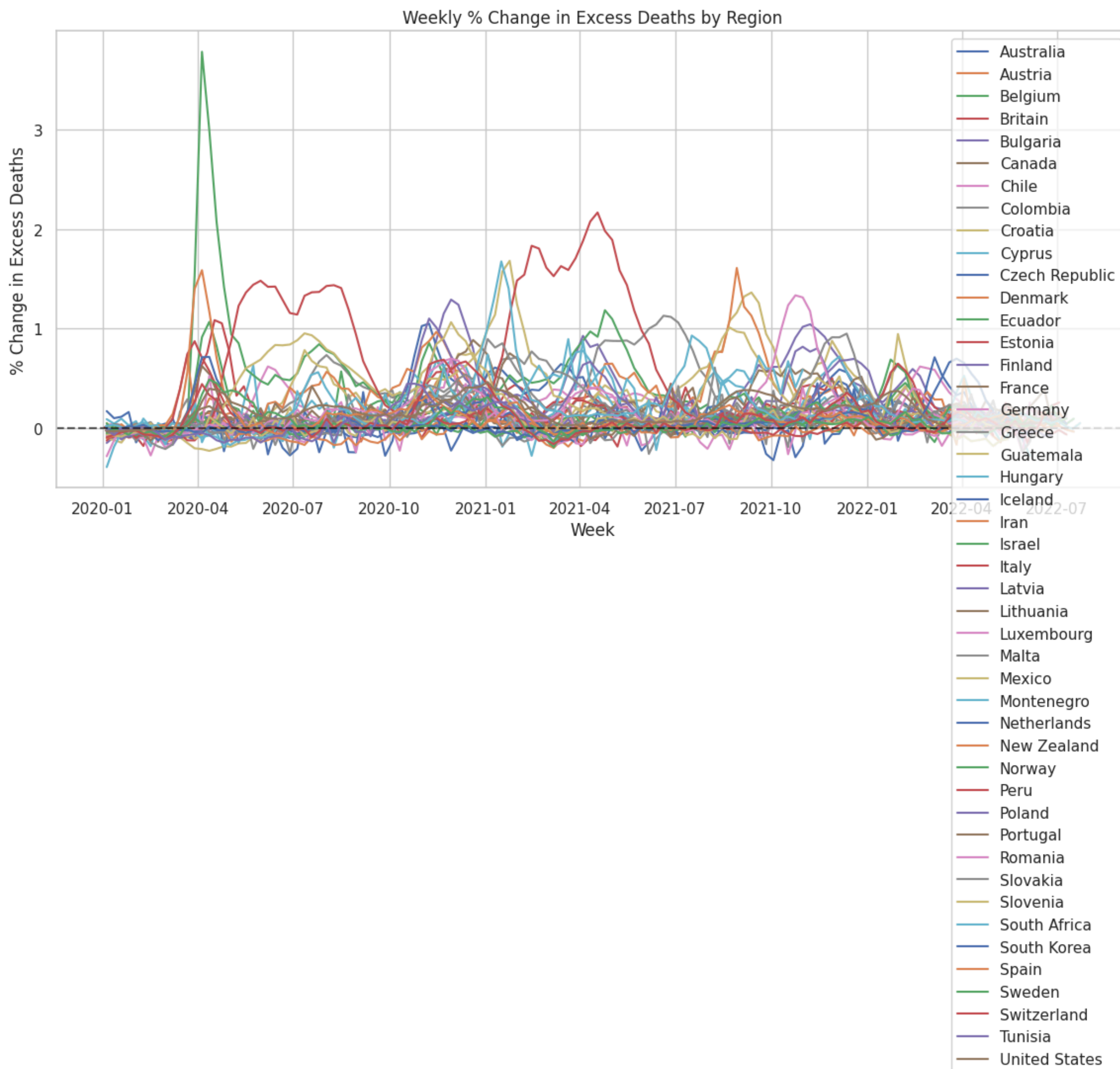
4. **Correlation Heatmap** – how different death metrics relate



5. **COVID vs Excess Deaths Rates** – bar chart comparison per region



6. **Excess Death % Change Over Time** – volatility and outbreak impact



🧠 Early Observations from First Few Rows

- The data for **Australia** in early 2020 shows **zero reported COVID-19 deaths** while some weeks had **negative excess deaths**, possibly due to lockdown effects reducing other causes of death (like traffic accidents).

- Values like `excess_deaths_per_100k` and `excess_deaths_pct_change` are already computed—saving preprocessing time.